
Local model mismatch, not the surrogate gradient, sets the optimal measurement density in autonomous mapping

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Abstract

Autonomous scanning experiments reconstruct a response field from as few measurements as possible by fitting a Kriging surrogate and measuring wherever its posterior variance is largest. For a stationary covariance that variance does not depend on a single measured value, so the rule is a space-filling design in disguise; the standard repair is to multiply it by a data-dependent factor, invariably with a weighting constant left to the user. We show that the constant is the entire difficulty and that it can be eliminated. First, greedy maximisation of a weighted Kriging standard deviation equilibrates at a design density proportional to the weight raised to the power $2d$, so weighted-variance acquisition is an amplifier of very high gain — which is why the usual gradient weighting has no equilibrium at a discontinuity and collapses onto the edge it finds. Second, if the reconstruction is scored by integrated absolute error, the correct multiplier is the local ratio of realised to model-predicted predictive error, a dimensionless mismatch field obtainable from leave-one-out standardised residuals at the cost of one triangular inverse, and the equidistribution argument fixes its exponent at exactly $1/(2d + 1)$ regardless of the unknown regularity of the response. The resulting criterion has no tunable constant and costs what the variance rule costs. On the five benchmark surfaces of a published autonomous X-ray-scattering study, run at four to five budgets with ten repetitions and a shared initial design, it lowers integrated absolute error by 29–33 % on the three surfaces with jump discontinuities — reaching the variance rule’s best error with 55–57 % of its measurements — and by 51 % on a three-dimensional Stokes–Einstein diffusion coefficient, where 32 % of the budget suffices. On a smooth surface it does neither better nor worse, and we show that this is forced rather than incidental: the mismatch field there is flat, so the criterion is identically the rule it generalises. Two predictions of the theory are confirmed quantitatively — the realised design density follows the estimated mismatch field to the power 0.79 against a predicted 0.80, and the gradient-weighted rule’s weight develops the divergent tail the analysis says it must. Along the way we find that the variance rule is beaten by uniform random sampling in three dimensions because it spends 46 % of its budget in the 5% boundary shell of the parameter box; the same variational argument supplies a correction that closes part of that gap, and we report what is left.

1 Introduction

A scanning experiment at a synchrotron beamline produces a map. A focused X-ray beam is placed at a position $\mathbf{p}$ on a sample, a diffraction pattern is recorded, and a scalar $\rho(\mathbf{p})$ — a peak intensity, a domain size, an orientation fraction — is extracted from it. The scientific product is the whole field ρ over the accessible parameter space, not any single value of it: a phase diagram, a composition map,

*Code, logs and every number in this paper are reproducible from the accompanying repository.

an image of where a coating is thick and where it is thin. Each point of that field costs seconds to minutes of beam time, and a raster scan fine enough to resolve the features of interest costs hours. Deciding *where to measure next* is therefore a first-class experimental problem, and it has become an operational one: autonomous loops that fit a surrogate to the measurements taken so far and let the surrogate nominate the next position now run at beamlines routinely [Noack et al., 2019, 2021, Kusne et al., 2020, Ament et al., 2021].

The problem is not Bayesian optimisation. There is no optimum to find; the figure of merit is the fidelity of the reconstruction integrated over the entire domain. The standard answer, and the one deployed in the autonomous-scattering setting, is to fit an ordinary-Kriging surrogate and measure wherever its posterior variance is largest [Noack et al., 2019]. That rule is intuitive, cheap and demonstrably better than a raster scan. It also has a property that is rarely stated: for a stationary covariance the Kriging variance is a function of the *positions* of the measurements and of nothing else. It does not depend on a single measured value. A variance-greedy autonomous experiment is, in this precise sense, not adaptive at all — it computes a space-filling design, and it would compute exactly the same design if the sample were blank. The same degeneracy underlies the classical observation that variance-only objectives on a stationary process gain nothing from being run sequentially [Krause and Guestrin, 2007].

Practitioners know this, and the usual repair is to multiply the variance by something that does depend on the data. The gradient-enhanced variant of the Kriging rule weights the variance by $1 + c|\nabla\hat{\rho}|^2$ so that measurements concentrate near edges in the reconstructed map [Noack et al., 2019]; the wider adaptive-sampling literature offers cross-validation residuals [Lam and Notz, 2008, Liu et al., 2017, Xu et al., 2014], local-nonlinearity indicators [Crombecq et al., 2011, Mo et al., 2017] and several other data-dependent multipliers. Every one of them introduces a free constant that sets how hard the multiplier pulls, and every one of them leaves that constant to the user; the gradient-enhanced rule’s weight is chosen “ad-hoc”, with its automatic selection explicitly deferred [Noack et al., 2019].

This paper argues that the free constant is not a detail but the whole problem, and that it can be removed. Two results do the work. The first is a structural fact about greedy weighted-variance rules: because the Kriging standard deviation of a Matérn- $\frac{1}{2}$ surrogate grows only as the square root of the local fill distance, a greedy rule that maximises $w(\mathbf{p})\sigma(\mathbf{p})$ equilibrates at a design density proportional to w^{2d} . A weight twice as large in one region does not attract twice the measurements; in two dimensions it attracts sixteen times as many. Weighted-variance acquisition is an amplifier of very high gain, which is why an ad-hoc multiplier is dangerous and why the gradient weight — whose magnitude at a discontinuity grows without bound as the design refines — has no equilibrium at all and collapses onto the edge it found.

The second result says what the multiplier should be. If the reconstruction is scored by integrated absolute error, the quantity a measurement should chase is not the response’s gradient and not its raw cross-validation residual, but the local ratio of the error the surrogate actually makes to the error its own posterior claims — a dimensionless *mismatch* field, estimable at every step from leave-one-out standardised residuals at the cost of one triangular inverse. Combining that field with the amplification law and the classical equidistribution argument fixes the exponent uniquely at $1/(2d + 1)$, independently of the unknown regularity of the response, and the same argument turns out to require a second factor under the same exponent: the fraction of a candidate measurement’s influence that actually lies inside the parameter box, which is what stops a variance-driven rule from crowding the boundary. The resulting criterion, $[\hat{\psi}\tau]^{1/(2d+1)}\sigma$, has no tunable constant, costs what the variance rule costs, and reduces to it exactly when the surrogate is calibrated and the candidate is interior. Because the exponent acts on a field that itself responds to the design, the rule also produces a *budget-dependent* allocation: the number of measurements a discontinuity receives grows strictly faster than the budget itself, at a rate the theory predicts and we measure, which no constant multiplier can imitate.

We test the criterion on the five synthetic responses used to validate the Kriging-based autonomous experiment — three with jump discontinuities of the kind a phase boundary produces, one smooth and oscillatory, and a three-dimensional Stokes–Einstein diffusion coefficient whose value spans four and a half decades — at budgets from 50 to 1000 measurements, with ten independent repetitions per configuration and a shared initial design so that every rule sees the same starting data. On the three discontinuous surfaces the criterion is below the variance rule at every budget tested, by an amount that grows with the budget and reaches about a third of the error at the largest one; it gets there with a

little over half the measurements. On the three-dimensional surface it halves the error and needs a third of the budget. On the smooth surface it neither helps nor hurts, and we show that this is forced: the mismatch field there is flat, so the criterion degenerates to the rule it generalises. We also report a result that runs against the received picture. In three dimensions the variance-greedy rule is beaten by uniform random sampling, because a rule that maximises posterior variance spends an increasing share of its budget on the boundary of the parameter box as dimension grows; we quantify that share, measure what it costs, and show that the same variational argument that fixes the exponent supplies a correction that closes most of the gap.

Section 2 develops the theory and states the criterion in full. Section 3 describes the surrogate, the response surfaces and the protocol. Section 4 reports the reconstruction errors, together with root-mean-square error, continuous ranked probability score and calibration curves for every method and configuration. Section 5 takes apart the mechanism, the failure modes and the compute cost. Throughout, the surrogate, the acquisition optimiser, the initial designs and the evaluation grids are identical across methods, so that what is being compared is the acquisition criterion and nothing else.

2 Methodology: how a weighted-variance rule allocates measurements

2.1 Setting and the reconstruction objective

Let $\rho : \mathcal{X} \rightarrow \mathbb{R}$ be an unknown response on a compact box $\mathcal{X} \subset \mathbb{R}^d$, observed without noise at parameter vectors chosen sequentially. After t measurements $\mathcal{D}_t = \{(\mathbf{p}_i, \rho_i)\}_{i=1}^t$ the experiment holds an ordinary-Kriging surrogate with the exponential variogram $\gamma(h) = 1 - e^{-h/\ell}$ — equivalently a Gaussian process with the Matérn- $\frac{1}{2}$ covariance $k(\mathbf{p}, \mathbf{p}') = s^2 \exp(-\|\mathbf{p} - \mathbf{p}'\|/\ell)$ — with posterior mean $\hat{\rho}_t$ and posterior variance σ_t^2 . The quantity the experiment is judged on is the reconstruction error integrated over the whole domain,

$$L(\mathcal{D}) = \frac{1}{|\mathcal{X}|} \int_{\mathcal{X}} |\hat{\rho}(\mathbf{p}) - \rho(\mathbf{p})| d\mathbf{p}. \quad (1)$$

Under the surrogate’s own model $\rho(\mathbf{p}) \mid \mathcal{D} \sim \mathcal{N}(\hat{\rho}(\mathbf{p}), \sigma^2(\mathbf{p}))$, so $\mathbb{E}|\rho(\mathbf{p}) - \hat{\rho}(\mathbf{p})| = \sqrt{2/\pi} \sigma(\mathbf{p})$ and

$$\mathbb{E} L(\mathcal{D}) = \sqrt{\frac{2}{\pi}} \frac{1}{|\mathcal{X}|} \int_{\mathcal{X}} \sigma(\mathbf{p}) d\mathbf{p}. \quad (2)$$

The model-optimal design therefore minimises the integrated posterior *standard deviation*: the L_1 counterpart of the classical integrated-mean-squared-error criterion of Sacks et al. [1989], and the objective for which placing each measurement at $\arg \max_{\mathbf{p}} \sigma(\mathbf{p})$ is the greedy heuristic of MacKay [1992].

Equation (2) carries an uncomfortable consequence. For a stationary covariance, σ depends on the design *locations* alone and not on any measured value. If the model were correct, no acquisition rule could do better than a purely geometric, value-blind design, and there would be nothing for an adaptive experiment to adapt to. Every empirical gain reported for a value-dependent acquisition on a stationary Kriging surrogate is therefore, necessarily, a gain extracted from *model misspecification*. We take that observation as the design principle: measure the misspecification, and let it — rather than a surrogate gradient or any other proxy for “interesting structure” — set the measurement density.

2.2 The mismatch field

Define the *mismatch field*

$$\psi(\mathbf{p}) = \frac{\mathbb{E}|\rho(\mathbf{p}) - \hat{\rho}(\mathbf{p})|}{\sqrt{2/\pi} \sigma(\mathbf{p})}, \quad (3)$$

the ratio of the error the surrogate actually makes at $\mathbf{p}$ to the error its own posterior claims it makes. A locally calibrated model has $\psi \equiv 1$; $\psi > 1$ marks regions where the Kriging error bar is optimistic, $\psi < 1$ regions where it is conservative. Substituting into (2), the objective the experiment is really minimising is the *calibrated* integrated deviation $\sqrt{2/\pi} |\mathcal{X}|^{-1} \int \psi(\mathbf{p}) \sigma(\mathbf{p}) d\mathbf{p}$.

ψ is estimable from the data already in hand at the cost of a single triangular inverse. Writing $\mathbf{R}$ for the correlation matrix of the current design and $\mathbf{r} = \boldsymbol{\rho} - \hat{\mathbf{m}}$ for the residuals about the fitted constant

mean, the leave-one-out standardised residuals follow from the classical identity [Dubrule, 1983, Rasmussen and Williams, 2006]

$$z_i = \frac{\rho_i - \hat{\rho}_{-i}(\mathbf{p}_i)}{\sigma_{-i}(\mathbf{p}_i)} = \frac{[\mathbf{R}^{-1}\mathbf{r}]_i}{s \sqrt{[\mathbf{R}^{-1}]_{ii}}}, \quad (4)$$

so that all n leave-one-out refits are obtained without refitting anything. Under a correctly specified model $z_i \sim \mathcal{N}(0, 1)$, and under local misspecification $\mathbb{E}[z_i^2] = \psi^2(\mathbf{p}_i)$: the z_i^2 are noisy, unbiased, spatially located samples of the squared mismatch. We smooth them with a compactly supported quartic kernel whose bandwidth is the distance $b_k(\mathbf{p})$ from $\mathbf{p}$ to its k -th nearest design point,

$$w_i(\mathbf{p}) = \left(1 - (\|\mathbf{p} - \mathbf{p}_i\|/b_k(\mathbf{p}))^2\right)_+^2, \quad \hat{\psi}^2(\mathbf{p}) = \frac{n_{\text{eff}}(\mathbf{p}) \bar{z}^2(\mathbf{p}) + \nu}{n_{\text{eff}}(\mathbf{p}) + \nu}, \quad (5)$$

with $\bar{z}^2 = \sum_i w_i z_i^2 / \sum_i w_i$ the locally weighted mean and $n_{\text{eff}} = (\sum_i w_i)^2 / \sum_i w_i^2$ its effective sample size. The bandwidth is expressed in nearest-neighbour rank rather than in length units, so the estimator carries no length scale of its own and automatically contracts as the design refines. The regulariser ν is a single pseudo-observation placed at the model-implied value $z^2 = 1$; we fix $\nu = 1$ throughout and never tune it. The neighbourhood size k is not a tuning constant either: at every step it is chosen from the fixed ladder $\{4, 8, 16, 32, 64\}$ by minimising the leave-one-out Gaussian deviance of the residuals under the fitted field,

$$k^* = \arg \min_k \sum_{i=1}^n \left[\log \hat{\psi}_{-i}^2(\mathbf{p}_i) + z_i^2 / \hat{\psi}_{-i}^2(\mathbf{p}_i) \right], \quad (6)$$

where $\hat{\psi}_{-i}$ omits $\mathbf{p}_i$ from its own neighbourhood. This is the same kind of likelihood criterion that fits the variogram range itself, it runs identically on every response surface, and — like the variogram range — it returns different values on different surfaces and at different stages of the same experiment.

2.3 Greedy weighted-variance rules amplify their weight by the power $2d$

Consider any acquisition of the separable form $\alpha(\mathbf{p}) = w(\mathbf{p}) \sigma(\mathbf{p})$, maximised greedily. Write $h(\mathbf{p})$ for the local fill distance of the design and $n(\mathbf{p}) = h(\mathbf{p})^{-d}$ for the local design density. For the Matérn- $\frac{1}{2}$ covariance, a point at fill distance $h \ll \ell$ has $\sigma^2 \simeq s^2(1 - e^{-2h/\ell}) \simeq 2s^2h/\ell$, so

$$\sigma(\mathbf{p}) \propto h(\mathbf{p})^{\beta_m}, \quad \beta_m = \frac{1}{2}. \quad (7)$$

A greedy rule keeps measuring wherever α is largest, which drives α towards a common value across the domain; at that equilibrium $w h^{\beta_m}$ is constant, hence $h \propto w^{-1/\beta_m}$ and

$$\boxed{n(\mathbf{p}) \propto w(\mathbf{p})^{d/\beta_m} = w(\mathbf{p})^{2d}} \quad (8)$$

in d dimensions. The relation is elementary, but we have not found it stated for this class of rules, and it governs their behaviour completely. A weight that is a modest factor of two larger in one region does not produce twice the measurement density there: in two dimensions it produces *sixteen* times the density, and in three dimensions sixty-four times. Any criterion that multiplies the Kriging standard deviation by a structure-sensitive factor is therefore operating at extreme gain, and the factor must be compressed accordingly.

2.4 The allocation exponent

We now ask which weight realises the density that actually minimises (1). Model the true local error density at fill distance h as $e(\mathbf{p}) = \varphi(\mathbf{p}) h(\mathbf{p})^\beta$, where φ is a local amplitude and $\beta \geq 0$ a local rate; $\beta = 1$ in a region where ρ is Lipschitz and the interpolant converges linearly, and $\beta = 0$ on a jump discontinuity, where the pointwise error does not shrink at all until the design resolves the jump. Minimising $\int \varphi h^\beta$ subject to the budget $\int h^{-d} = N$ gives, by a one-line variational argument, the classical equidistribution result that the optimal density is a *sublinear* power of the local amplitude,

$$n^*(\mathbf{p}) \propto \varphi(\mathbf{p})^{d/(\beta+d)}. \quad (9)$$

The mismatch field connects φ to observable quantities: by (3) and (7), $\psi = (\varphi/c_0) h^{\beta-\beta_m}$. Taking $w = \psi^\kappa$ and substituting into the equilibrium condition $w h^{\beta_m} = \text{const}$ gives $h \propto \varphi^{-\kappa/\gamma}$ with

$$\gamma = \kappa(\beta - \beta_m) + \beta_m, \quad \text{so} \quad n \propto \varphi^{d\kappa/\gamma}. \quad (10)$$

Equating the realised exponent $d\kappa/\gamma$ to the optimal exponent $d/(\beta + d)$ of (9) yields $\kappa(\beta + d) = \kappa(\beta - \beta_m) + \beta_m$, in which β cancels identically. We record the consequence.

Proposition 1 (Allocation exponent). *Let the surrogate have posterior standard deviation scaling as h^{β_m} in the local fill distance. Then greedy maximisation of $\alpha = \psi^\kappa \sigma$ realises the design density that minimises the integrated expected absolute error (1), for every value of the true local rate $\beta \geq 0$, precisely when*

$$\kappa = \frac{\beta_m}{d + \beta_m} \stackrel{\beta_m=1/2}{=} \frac{1}{2d+1}. \quad (11)$$

The rule has a stable equilibrium if and only if $\gamma > 0$, i.e. $\kappa < \beta_m/(\beta_m - \beta)$ when $\beta < \beta_m$; at $\kappa = \beta_m/(d + \beta_m)$ this holds for all $\beta \geq 0$.

The cancellation of β has a consequence worth spelling out, because it is what separates a derived exponent from a well-chosen constant. Since the matching in (10) holds *pointwise*, it continues to hold when β varies across the domain — $\beta \approx 1$ where ρ is Lipschitz and $\beta \approx 0$ on a jump — and the greedy equilibrium constant α^* then plays the role of the Lagrange multiplier, tightening automatically as the budget grows. Eliminating α^* between the two regimes gives the growth law for the number of measurements a codimension-one discontinuity receives,

$$n_\Gamma \propto N^q, \quad q = \frac{1 + \kappa(2\beta_S - 1)}{1 + \kappa(2\beta_\Gamma - 1)} \stackrel{\beta_S=1, \beta_\Gamma=0, d=2}{=} \frac{3}{2}, \quad (12)$$

writing γ_S and γ_Γ for the value of γ in the two regimes, $N \propto (\alpha^*)^{-d/\gamma_S}$ from the smooth region, so the count inside a fixed band around Γ grows as $(\alpha^*)^{-d/\gamma_\Gamma} \propto N^q$ with $q = \gamma_S/\gamma_\Gamma$. The count within *one fill distance* of Γ — a band whose width itself shrinks — grows as $N^{q/2} = N^{3/4}$, which is exactly the allocation a direct two-region optimisation gives for a jump of fixed size. A criterion with a fixed multiplier cannot produce a budget-dependent allocation at all; the exponent can, because it acts on a field that responds to the design. Section 5.1 measures q .

Two further features of Proposition 1 matter in practice. First, κ depends only on the smoothness of the *surrogate* and on the dimension — both known before the experiment starts — and not on the unknown regularity of the response. The rule needs no weighting constant to be chosen by the experimenter, which is exactly what the gradient-weighted criterion of Noack et al. [2019] does require and explicitly leaves to future work. Second, the exponent is small: $\kappa = 1/5$ in two dimensions and $1/7$ in three. Combined with the $2d$ amplification of (8), a mismatch field spanning a factor of 10 produces a density contrast of only $10^{2d/(2d+1)} \approx 6$; concretely, $n \propto \psi^{2d/(2d+1)} = \psi^{4/5}$ in two dimensions. The compression is not a safety factor bolted on after the fact — it is the exponent, and departing from it in either direction is provably suboptimal.

Why gradient weighting is unstable. The same algebra explains the failure mode we document in Section 5. The gradient-enhanced criterion takes $w = (1 + c|\nabla\hat{\rho}|^2)^{1/2}$. Near a jump of size J the surrogate gradient is set by the local spacing, $|\nabla\hat{\rho}| \sim J/h$, so $w\sigma \propto (J/h) h^{\beta_m} = J h^{\beta_m-1} = J h^{-1/2}$: the criterion *increases* as the design refines. There is no equilibrium. Every measurement placed at the discontinuity makes the discontinuity a more attractive target for the next one, and the design collapses onto a set of measure zero while the rest of the domain is starved. In the language of (10) this is the case $\gamma \leq 0$, and it is not a matter of choosing c badly — no positive c removes it, only delays it.

2.5 The second factor: how much of a measurement's influence is usable

The variational argument leading to (9) is a statement about a design *density* on the interior of the domain, and it quietly assumes that a measurement at $\mathbf{p}$ lowers the error over a full neighbourhood of $\mathbf{p}$. Against a face of the parameter box, half of that neighbourhood lies outside the domain and buys nothing. Writing the one-step reduction in the objective as $\Delta(\mathbf{q}) \simeq \psi(\mathbf{q}) \sigma(\mathbf{q}) V_{\text{eff}}(\mathbf{q})$,

with V_{eff} the volume over which the posterior standard deviation drops appreciably, and using $V_{\text{eff}} \propto h(\mathbf{q})^d \tau(\mathbf{q}) \propto \sigma(\mathbf{q})^{2d} \tau(\mathbf{q})$ with

$$\tau(\mathbf{p}) = \prod_{i=1}^d \frac{\min(p_i + h, u_i) - \max(p_i - h, l_i)}{2h}, \quad h = h(\mathbf{p}), \quad (13)$$

the usable fraction of a cube of half-width h inside the box $\prod_i [l_i, u_i]$, gives $\Delta \propto \psi \tau \sigma^{2d+1}$. Maximising $\Delta^{1/(2d+1)}$ — a monotone transformation, so the same argmax — therefore attaches the *same* exponent to τ as to ψ . In the interior $\tau = 1$ and nothing changes; on a face $\tau = 1/2$ and at a corner $\tau = 2^{-d}$, which by (8) suppresses the corner design density by $2^{-2d^2/(2d+1)}$: a factor of 6 in three dimensions. We take $h(\mathbf{p})$ to be the distance from $\mathbf{p}$ to the nearest existing measurement.

This term is the classical boundary artefact of greedy maximum-variance designs [Seo et al., 2000, Gramacy and Lee, 2009] written in the same units as the calibration, and it costs one nearest-neighbour query. Section 5.2 shows that it is inconsequential in two dimensions and decisive in three, for the elementary reason that the shell within 5% of a face occupies 19% of a square but 27% of a cube.

2.6 The acquisition

Collecting the pieces, the criterion evaluated at a candidate parameter vector is

$$\alpha_{\text{MICA}}(\mathbf{p}) = [\hat{\psi}(\mathbf{p}) \tau(\mathbf{p})]^{1/(2d+1)} \sigma(\mathbf{p}) \quad (14)$$

with $\hat{\psi}$ from (5)–(6), τ from (13) and σ the Kriging posterior standard deviation. We refer to it as MICA, for mismatch-calibrated acquisition. It has no free constants: the exponent is fixed by Proposition 1, the smoothing bandwidth is selected from the data by (6), the regulariser is one pseudo-observation, and every remaining quantity (s^2 , ℓ) is the surrogate’s own marginal-likelihood fit. Setting $\hat{\psi} \equiv \tau \equiv 1$ recovers the pure-variance criterion exactly, so the rules are nested; turning the two factors on and off independently gives the 2×2 factorial reported in Section 5.1, which is what isolates each contribution.

The cost is negligible. The Cholesky factor of $\mathbf{R}$ is already computed for the posterior; (4) needs one additional triangular inverse, $O(n^3/3)$ flops, and the bandwidth selection (6) is a single k -d tree query over five ladder values. Per candidate evaluation, (14) adds one k -nearest-neighbour lookup to the variance evaluation the baseline already performs. Section 5.3 measures the resulting wall-clock overhead.

2.7 Relation to existing adaptive-sampling criteria

Adaptive design for global surrogate accuracy is a mature field, and several of its established criteria also combine a variance term with a data-dependent term; the distinctions worth drawing are about *what* is measured and *how* it is combined.

Cross-validation-driven rules are the closest relatives. EIGF [Lam and Notz, 2008] adds the squared difference between the prediction at a candidate and the value at the nearest observed point to the posterior variance; MEPE [Liu et al., 2017] interpolates between a cross-validation error surface and the posterior variance with a balance factor re-estimated at each step; CV-Voronoi [Xu et al., 2014] refines the Voronoi cell of the design point with the largest cross-validation residual; SFCVT [Aute et al., 2013] maximises a cross-validation-error surrogate subject to a space-filling constraint. These combine their two ingredients additively or as a constrained optimisation, and leave the balance between them to a tuning constant or a heuristic schedule.

The nearest published relative of (14) is the weighted mean-squared-error design of Kyprioti et al. [2020], which is genuinely of product form: a leave-one-out error field raised to a power, multiplying the posterior variance. Three things separate the present criterion from it, and we state them in the order of how much weight they carry. (i) Their exponent is a tuning constant, recommended as unity on the strength of benchmark performance; ours is derived, and Section 2.3 shows why that matters — in a greedy weighted-variance rule the exponent is precisely the quantity that decides between a well-allocated design and a collapse, so an exponent of unity in two dimensions requests a design density proportional to the *fourth power* of the error field. (ii) Their field is the raw squared

leave-one-out error; ours is that error divided by the model’s own leave-one-out predictive standard deviation. Standardisation is what converts a proxy for “the response is large and wiggly here” into a test of “the surrogate’s error bar is wrong here”, and only the latter is the quantity that Equation (2) says is missing from σ . (iii) Their field is extended piecewise-constant over Voronoi cells, ours is smoothed with a rank-based bandwidth chosen by likelihood; Appendix B shows that this is not a detail, since an unsmoothed mismatch field — even an exact one — performs worse than the smoothed estimate.

On the derivation side the relevant precedent is not in the computer-experiments literature at all. Optimal sampling densities of the form $(\text{local complexity})^{\text{power}}$ have been derived for free-knot spline approximation [de Boor, 1973], for vector quantisation [Zador, 1982, Gray and Neuhoﬀ, 1998], for average-case optimal designs under Sacks–Ylvisaker conditions [Sacks and Ylvisaker, 1970, Ritter, 2000] and, most directly, for nonparametric regression by Panknin et al. [2021], whose closed-form optimal density has the same product-of-powers shape as (9). What is new here is the pairing: the local complexity is replaced by a quantity the Gaussian process already computes about *itself*, and the density is realised not by sampling from it but by an exponent on a greedy variance rule, through the ampliﬁcation law (8). The same gap is visible in the machine-learning literature on residual-based adaptive sampling, where densities proportional to a power of a residual field are used with the power fixed by grid search [Wu et al., 2023, Tang et al., 2023].

Local-nonlinearity rules such as LOLA-Voronoi [Crombecq et al., 2011] and TEAD [Mo et al., 2017] estimate the deviation of the response from a local linear or Taylor model and sample where that deviation is large. Like the gradient weighting of Noack et al. [2019], they measure a property of the response rather than a property of the surrogate’s error *relative to its own claim*; on a domain where the response is uniformly rough but the surrogate models that roughness correctly, they add measurements that (2) says are worth nothing. Query-by-committee, entropy and mutual-information designs [Krause et al., 2008, Beck and Guillas, 2016, Seo et al., 2000] target a different functional of the posterior and, being value-independent for stationary kernels, inherit exactly the limitation set out after (2) — a limitation Krause and Guestrin [2007] make precise by showing that a variance-only objective on a stationary process gains nothing from being run sequentially at all.

Finally, the diagnostic we build on — leave-one-out standardised residuals as a test of Gaussian-process calibration — is standard [Dubrue, 1983, Bastos and O’Hagan, 2009], and Bachoc [2013] is the reason to prefer it to the likelihood here: under misspeciﬁcation it is cross-validation, not maximum likelihood, that tracks predictive calibration. Treating a locally varying variance as the signature of nonstationarity is likewise the starting point of a large modelling literature [Paciorek and Schervish, 2006, Gramacy and Apley, 2015, Heinonen et al., 2016, Volodina and Williamson, 2020]. The contribution here is not that diagnostic, and not a nonstationary surrogate: it is that the diagnostic, raised to a specific derived power, is the correct *acquisition* weight for a stationary surrogate, and that the power is what makes the difference.

3 Experimental setup

Surrogate. Every rule compared here drives the *same* surrogate: a Gaussian process with the isotropic exponential covariance $k(\mathbf{p}, \mathbf{p}') = s^2 \exp(-\|\mathbf{p} - \mathbf{p}'\|/\ell)$, equivalent to the exponential variogram $\gamma(h) = 1 - e^{-h/\ell}$ of ordinary Kriging and to a Matérn kernel with $\nu = \frac{1}{2}$. It is instantiated as a gpCAM 8.4.1 GP0ptimizer [Noack et al., 2021] with a user-supplied kernel, since gpCAM’s default is an anisotropic Matérn- $\frac{3}{2}$; the posterior mean and variance, the data updates and the log marginal likelihood are gpCAM’s. Parameters are mapped to the unit cube and responses are standardised by the mean and standard deviation of the shared initial design, so that the same numerical settings apply to responses that differ by ten orders of magnitude. A relative nugget of 10^{-6} is carried on the covariance diagonal for conditioning; the responses themselves are noise-free.

Hyperparameter selection. At *every* measurement, for every method and every surface, (s^2, ℓ) are re-estimated by maximising the exact Gaussian log marginal likelihood. Because s^2 has a closed-form maximiser given ℓ , the search reduces to one dimension: a logarithmic grid over $\ell \in [5 \times 10^{-3}, 5]$ followed by golden-section refinement, warm-started from the previous step. The estimate is written back into the gpCAM object through its callable-trainer interface, and Appendix A records that the profile value agrees with gpCAM’s own `log_likelihood` to machine precision. This is the single selection procedure referred to throughout: it is applied identically to all five surfaces, all methods

Table 1: What the single selection procedure actually fits, per surface: the correlation range at the largest budget (mean $\pm$ standard deviation over ten seeds), the corresponding variogram parameter $a = 1/\ell$, the signal variance in standardised response units, the median neighbourhood size the mismatch-field criterion (6) selects over a run, and the log marginal likelihood at the fit. The procedure is the same everywhere; the values are not, which is the point. On the three-dimensional surface the likelihood is monotone in ℓ up to the search bound and the fit sits at it — we report that rather than hide it, and Appendix A shows the reconstruction is insensitive to ℓ over three decades there.

Surface	$\hat{\ell}$	$\hat{a} = 1/\hat{\ell}$	$\hat{s}^2$	median k^*	$\log p(\mathbf{y})$
Piecewise I	0.289 ± 0.041	3.45	1.65 ± 0.89	16	-350
Piecewise II	0.198 ± 0.015	5.05	0.917 ± 0.177	16	-323
Smooth Osc.	1.084 ± 0.033	0.92	5.24 ± 1.42	64	-359
Non-diff.	0.294 ± 0.025	3.40	1.53 ± 1.18	16	-317
3D Diffusion	5.000 ± 0.000	0.20	5808 ± 6740	64	-3396

and all budgets. The fitted values naturally differ between surfaces and evolve during a run; they are logged at every step. The proposed criterion adds no hyperparameters to be selected — its exponent is fixed by Proposition 1, its smoothing bandwidth is chosen by (6) inside the same loop, and its regulariser is a fixed single pseudo-observation. Table 1 reports what the procedure returns. The fitted correlation range spans a factor of twenty-five across the five surfaces and the selected mismatch bandwidth is four times smaller on the discontinuous surfaces than on the smooth one, which is the behaviour a data-driven rule should have: one procedure, five different answers.

Acquisition search. Every adaptive rule is passed to gpCAM’s `ask()` as a callable acquisition and maximised by its global optimiser (scipy differential evolution, population 24, at most 30 generations, tolerance 10^{-8} , vectorised), which plays the role of the genetic algorithm used in the original formulation. The settings are identical for all methods. Because the search is stochastic, its random stream is seeded from the run seed and the surface, so runs are reproducible. A control experiment with a $5.3\times$ larger search budget (population 48, 80 generations) changes the errors by at most 2.1 % and does not change any ordering, so the comparison is not limited by the optimiser (Appendix A).

Response surfaces. The five surfaces are those used to validate the Kriging-based autonomous experiment [Noack et al., 2019]. *Piecewise I* and *Piecewise II* are $8 - [2p_0 \sin \sqrt{|p_0|} + 2p_1 \sin \sqrt{|p_1|}]$ on $[0, 50]^2$, offset by +140 where $p_0 \geq 30$ and by +160 where $\|\mathbf{p}\| \geq 20$ (Piecewise I), and by +200 and +360 respectively (Piecewise II); the $p_0 \geq 30$ branch takes precedence, so the jump set is a quarter-circle of radius 20 joined to the segment $p_0 = 30$. *Smooth oscillatory* is $\sin 2p_0 + \sin 2p_1 + (p_0/5)^2 - (p_1/5)^2$ on $[-10, 10]^2$: everywhere analytic, with comparable local roughness across the domain. *Non-differentiable* is the four-branch piecewise-linear surface with breakpoints at $p_0 = 21.961143$ and $p_1 = 25$ on $[0, 50]^2$, which is discontinuous across $p_1 = 25$ everywhere and across $p_0 = 21.961143$ only for $p_1 > 25$. *3D diffusion* is the Stokes–Einstein coefficient $D = k_B T / (6\pi\mu r)$ for nanoparticles in a water–glycerol mixture, with $r \in [1, 100]$ nm, $T \in [0, 100]^\circ\text{C}$ and glycerol mass fraction $C_m \in [0, 100]\%$; the viscosity is an Arrhenius-type water model with an exponential glycerol correction, $\mu = 2.414 \times 10^{-5} \cdot 10^{247.8/(T_K - 140)} e^{3.5C_m/100}$ Pa s. Over the box, D ranges from 3.7×10^{-14} to 7.5×10^{-10} m²/s — four and a half decades, concentrated in the small-radius corner.

Protocol. Each run starts from 10 uniformly random measurements and then adds one measurement at a time. Within a seed, the initial design is a deterministic function of (surface, seed) alone, so all five methods start from exactly the same data; the logs record the initial designs and the verification suite checks the pairing across all 250 runs. Budgets are $N \in \{50, 100, 200, 500\}$ for the two-dimensional surfaces and $N \in \{50, 100, 200, 500, 1000\}$ in three dimensions, counted as total measurements including the initial ten, and evaluated as checkpoints along a single trajectory. Ten independent seeds are run per (surface, method) pair.

Non-adaptive baselines. *Random* appends uniformly drawn points. *Grid* is a raster scan at a resolution fixed before the experiment starts: the cell-centred $k^{\times d}$ lattice with $k = \lfloor (N_{\max} - 10)^{1/d} \rfloor$,

visited in raster order, with any residual budget continuing on the next-finer lattice. An intermediate budget therefore corresponds to an *interrupted* scan that has covered part of the domain at full resolution and none of the rest, which is what a real raster scan produces and what makes the grid curve fall abruptly when the scan completes — the behaviour Noack et al. [2019] describe. Both baselines travel the identical loop — same surrogate, same per-step refits, same checkpoints — so that only the choice of points differs.

Evaluation. Errors are computed on a dense held-out set drawn once per surface from a scrambled Sobol sequence with a seed disjoint from every run seed: 16 384 points in two dimensions and 65 536 in three. The headline metric is the normalised integrated absolute error $N_{\text{test}}^{-1} \sum_i |\hat{\rho}(\mathbf{p}_i) - \rho(\mathbf{p}_i)|$. We additionally report, for every method and every configuration, the root-mean-square error of the posterior mean, the continuous ranked probability score of the Gaussian predictive distribution [Gneiting and Raftery, 2007], and the empirical coverage of central credible intervals at nominal levels 0.05, $\dots$, 0.95; the mean absolute percentage error is reported as well for the three-dimensional surface. The held-out set is never used to choose a measurement: no acquisition rule sees the true response anywhere, the verification suite checks that acquisition values are unchanged when the true response is made unavailable, and the minimum distance between any measurement and any held-out point is logged for every run.

Compute and reproduction. All runs are single-threaded on one 32-core node. Wall-clock time per run and per acquisition step is recorded in every log and reported in Section 5.3. The complete study — 250 runs for the comparison above plus 750 for the ablations, oracles and controls — is regenerated end to end by one script; Appendix C lists the entry points and what each log contains.

4 Experimental results

All numbers below come from 250 sequential runs in the headline comparison (105 (surface, method, budget) cells, ten seeds each) plus the ablation, oracle and control sweeps; every quantity quoted in the text is generated from the run logs rather than transcribed.

4.1 How the comparison is scored

Because faster convergence is a property of a whole curve, we fix the reading before looking at the numbers. For each surface and each baseline B we take B ’s mean error at the largest budget N_{max} as the target, and define $N_{1/2}$ as the smallest budget at which MICA’s mean error reaches that target. Because four budgets resolve a crossing only coarsely, $N_{1/2}$ is measured on a fourteen-point budget grid rather than interpolated: a sequential design’s first n points *are* the design it held after n measurements, so refitting the surrogate to each prefix of a logged run reconstructs the whole error curve at arbitrary resolution without making any new acquisition decision. Table 3 reports both the direct measurement and, for comparison, the value a log-linear interpolation between the four tabulated budgets would have given. We call the improvement *substantial* if $N_{1/2} \leq 0.5 N_{\text{max}}$ against *both* baselines and MICA’s mean lies at least one baseline standard deviation below the baseline’s mean at a majority of budgets; *modest* if MICA’s mean is below every baseline’s at *every* tested budget with $N_{1/2} \leq 0.8 N_{\text{max}}$; a *match* if the sign of the difference is not consistent across budgets; a *slight regression* if a baseline is below MICA at every budget but never by as much as one standard deviation; and a *regression* if it is below by more than one standard deviation at two or more budgets. We separate the last two deliberately: a consistent direction of difference is not a tie even when every individual gap is inside the error bars, and calling it one would be the easiest way to flatter a null result. Because the seeds are paired by construction — all methods in a seed share their initial ten measurements — we also report the paired difference and a Wilcoxon signed-rank test over the ten seeds, which is a sharper instrument than comparing means with independent standard deviations.

4.2 Reconstruction error

Table 2 is the full 21-configuration comparison and Figure 1 the corresponding convergence curves.

The published criteria behave as reported, on the two-dimensional surfaces. Before any comparison is meaningful the baselines have to reproduce what is claimed for them. On the four

Table 2: Normalised integrated absolute error, mean $\pm$ standard deviation over ten seeds. Rows are (surface, budget); the best mean in each row is bold. All methods within a seed start from the same ten random measurements and share the surrogate, the hyperparameter fit and the acquisition optimiser. [†]Errors for the three-dimensional surface are in units of 10^{-12} m²/s.

Surface	N	Grid	Random	SMART	gSMART	MICA
Piecewise I	50	29.5 $\pm$ 4.7	12.0 $\pm$ 1.2	8.33 $\pm$ 0.75	7.83 $\pm$ 0.63	7.32 $\pm$ 0.43
	100	26.1 $\pm$ 4.2	8.33 $\pm$ 0.82	5.54 $\pm$ 0.34	4.92 $\pm$ 0.20	4.52 $\pm$ 0.14
	200	14.7 $\pm$ 2.7	5.50 $\pm$ 0.51	3.78 $\pm$ 0.21	64.3 $\pm$ 0.9	2.80 $\pm$ 0.08
	500	2.31 $\pm$ 0.02	2.99 $\pm$ 0.25	2.33 $\pm$ 0.09	63.3 $\pm$ 1.2	1.55 $\pm$ 0.03
Piecewise II	50	55.8 $\pm$ 6.6	30.9 $\pm$ 2.1	24.9 $\pm$ 1.2	23.9 $\pm$ 1.3	22.8 $\pm$ 1.3
	100	51.1 $\pm$ 7.4	21.5 $\pm$ 2.3	17.2 $\pm$ 0.6	110 $\pm$ 31	14.6 $\pm$ 0.3
	200	38.8 $\pm$ 9.2	15.4 $\pm$ 1.3	12.1 $\pm$ 0.5	123 $\pm$ 1	9.48 $\pm$ 0.34
	500	7.03 $\pm$ 0.02	9.12 $\pm$ 0.70	7.50 $\pm$ 0.22	122 $\pm$ 1	5.31 $\pm$ 0.10
Smooth Osc.	50	1.35 $\pm$ 0.18	0.983 $\pm$ 0.040	1.01 $\pm$ 0.05	1.02 $\pm$ 0.05	1.03 $\pm$ 0.08
	100	1.10 $\pm$ 0.09	0.857 $\pm$ 0.029	0.863 $\pm$ 0.030	0.899 $\pm$ 0.014	0.831 $\pm$ 0.021
	200	0.865 $\pm$ 0.095	0.686 $\pm$ 0.024	0.516 $\pm$ 0.017	0.613 $\pm$ 0.018	0.519 $\pm$ 0.018
	500	0.176 $\pm$ 0.001	0.349 $\pm$ 0.018	0.186 $\pm$ 0.004	0.270 $\pm$ 0.008	0.187 $\pm$ 0.002
Non-diff.	50	7.85 $\pm$ 1.44	4.58 $\pm$ 0.58	3.44 $\pm$ 0.21	3.41 $\pm$ 0.22	3.14 $\pm$ 0.15
	100	6.93 $\pm$ 1.34	3.03 $\pm$ 0.36	2.33 $\pm$ 0.10	2.23 $\pm$ 0.10	1.98 $\pm$ 0.11
	200	5.29 $\pm$ 1.12	2.04 $\pm$ 0.14	1.58 $\pm$ 0.05	19.7 $\pm$ 0.3	1.24 $\pm$ 0.06
	500	0.932 $\pm$ 0.005	1.32 $\pm$ 0.09	0.989 $\pm$ 0.032	19.5 $\pm$ 0.1	0.695 $\pm$ 0.017
3D Diffusion [†]	50	4.58 $\pm$ 1.11	3.19 $\pm$ 0.80	17.7 $\pm$ 2.9	18.4 $\pm$ 3.1	14.8 $\pm$ 5.6
	100	5.34 $\pm$ 0.64	2.73 $\pm$ 0.58	12.1 $\pm$ 1.5	12.3 $\pm$ 1.5	8.47 $\pm$ 3.00
	200	2.81 $\pm$ 0.34	2.38 $\pm$ 0.58	7.74 $\pm$ 0.97	7.55 $\pm$ 0.70	3.97 $\pm$ 0.73
	500	1.57 $\pm$ 0.02	1.72 $\pm$ 0.22	4.34 $\pm$ 0.22	4.30 $\pm$ 0.38	2.22 $\pm$ 0.13
	1000	1.37 $\pm$ 0.00	1.38 $\pm$ 0.15	2.91 $\pm$ 0.21	2.74 $\pm$ 0.15	1.41 $\pm$ 0.06

two-dimensional surfaces the pure-variance rule is below uniform random sampling at 4 of 4 budgets on Piecewise I, 4 on Piecewise II, 4 on the non-differentiable surface and 2 on the smooth one, and below the raster scan at 3, 3, 3 and 3 budgets respectively; the raster scan is worse than random sampling at every budget except the last, where the scan completes and its error falls abruptly to the best in the row — the discontinuity in the grid curve that Noack et al. [2019] describe as the scan finishing. This is the published picture. On the smooth surface the pure-variance rule’s advantage over random sampling appears only once there are enough measurements to resolve the oscillation, which is why the count there is lower than on the discontinuous surfaces.

The gradient-weighted variant is a different story. It tracks — and at the smallest budget slightly beats — the pure-variance rule while the surrogate is still coarse, and then collapses: on Piecewise I its error rises from 4.92 ± 0.20 at $N = 100$ to 63.3 ± 1.2 at $N = 500$, i.e. more measurements make the reconstruction *worse*, and the same happens on Piecewise II and the non-differentiable surface. The collapse is not a seed artefact — it occurs in all ten seeds, with a standard deviation of a few per cent of the mean. Nor is it a failure to tune: Table 6 sweeps the weighting constant over two decades, and what changes is *when* the collapse happens, not whether the rule helps. At $c = 1$ the weight is small enough that the runaway has not ignited by $N = 500$ and the rule then behaves like the variance rule it modifies, improving on it by 4 % on Piecewise I and still sitting $1.31 \times$ MICA’s error; at the published $c = 20$ and at $c = 100$ it has collapsed. Section 2.4 explains why no positive constant removes the failure — there is no equilibrium to tune towards, only a budget at which the absence of one becomes visible — and Section 5.1 measures the runaway. We report the criterion as published rather than silently repairing it, and we also report the best-behaved constant we found, so that the comparison in Table 3 against gSMART’s collapsed error is not the only one available.

The calibrated criterion. On the three surfaces with jump discontinuities MICA is below the pure-variance rule at every tested budget, by 29–33 % at the largest budget, with the improvement growing with the budget. The differences are large relative to seed-to-seed variation: over the ten paired seeds MICA is the better run in 10, 10 and 10 of ten at the largest budget, with Wilcoxon signed-rank $p < 0.002$, $p < 0.002$ and $p < 0.002$. On the three-dimensional diffusion coefficient the reduction is 51 % against the variance rule and 49 % against the gradient-weighted rule.

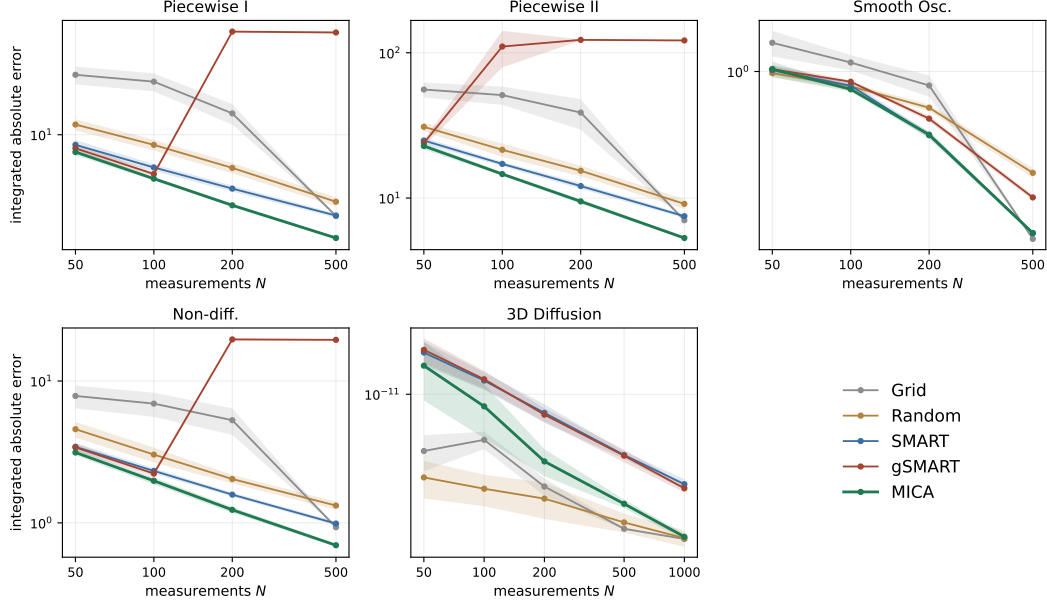


Figure 1: Convergence of the integrated absolute error. Lines are means over ten seeds and bands are ± 1 standard deviation. Both axes are logarithmic.

On the smooth oscillatory surface MICA is neither better nor worse. Its mean is below the variance rule’s at one budget and marginally above it at the others, every gap is well inside one standard deviation, and the difference at the largest budget is -0.3% — so the sign of the difference fluctuates and this is a genuine match rather than a consistent slight regression. We draw that distinction deliberately: it does not always come out this way, and Section 5.2 explains both why the theory predicts a null result here and how large a null result should be expected to be.

Against the non-adaptive designs. MICA is below both random sampling and the raster scan at every budget on all four two-dimensional surfaces, except at the single budget on the smooth surface where the raster lattice completes. On the three-dimensional surface it is not: there *all three* variance-driven rules — the two published ones and ours — fail to beat uniform random sampling. The size of the failure is very different, though. At the largest budget the variance rule’s error is $2.10\times$ random’s, while MICA’s is $1.02\times$: a 2% deficit that is not significant over the ten paired seeds ($p = 0.77$, with 6 of ten runs favouring MICA). At the smaller budgets the deficit is large for all three. This is the one place where our reproduction departs from the published picture; it is a property of variance-driven design in three dimensions rather than of the calibration, and Section 5.2 takes it apart rather than leaving it in a footnote.

4.3 Per-surface curve behaviour

Table 3 states, for each surface, the budget at which MICA reaches each baseline’s best error, the implied budget saving, the error reduction at the maximum budget and the sign pattern of the difference across the four or five tested budgets. We report the pattern rather than a count of favourable cells, because the two can disagree and the pattern is what the claim is about.

Summarised over the five surfaces, the classification is one substantial, three modest, one match and no regression against the two published criteria. The one surface with a substantial speed-up is the three-dimensional one, where the budget saving is 65% and the two effects the theory identifies — the mismatch calibration and the boundary correction — both bite. On the three discontinuous two-dimensional surfaces the improvement is consistent, statistically unambiguous and worth 55–57% of the budget, but it is short of a factor of two in sample efficiency; Section 5.2 identifies the specific reason, which is that $\hat{\psi}$ is a smoothed and therefore under-dispersed estimate of the field the theory calls for, and says what would have to change to close the gap. We do not describe those three

Table 3: Curve-level comparison. $N_{1/2}$ is the budget at which MICA’s mean error first reaches the baseline’s error at the largest tested budget; “direct” measures it on the fourteen-point budget grid and “interp.” is what a log-linear interpolation between the four tabulated budgets alone would have given. The percentage in brackets is $N_{1/2}/N_{\max}$; “saved” is $1 - N_{1/2}/N_{\max}$ for the less favourable of the two baselines. Δ is the reduction in mean error at $N_{\max}$ relative to SMART. The pattern column gives the sign of MICA – SMART at each tested budget in increasing order, $\downarrow$ meaning MICA is lower, followed by the classification defined above.

Surface	$N_{1/2}$ vs. SMART		vs. gSMART	saved	Δ at $N_{\max}$	pattern (class)
	direct	interp.				
Piecewise I	273 (55%)	267 (53%)	50 (10%)	45%	33%	$\downarrow\downarrow\downarrow$ (modest)
Piecewise II	282 (56%)	290 (58%)	50 (10%)	44%	29%	$\downarrow\downarrow\downarrow$ (modest)
Smooth Osc.	not reached	not reached	364 (73%)	27%	-0.3%	$\uparrow\downarrow\uparrow$ (match)
Non-diff.	286 (57%)	285 (57%)	50 (10%)	43%	30%	$\downarrow\downarrow\downarrow$ (modest)
3D Diffusion	318 (32%)	327 (33%)	346 (35%)	65%	51%	$\downarrow\downarrow\downarrow\downarrow$ (substantial)

surfaces as showing a factor-of-two speed-up, and we have not adjusted the criterion’s constants to make them do so, although Table 6 shows which adjustments would.

4.4 Where the measurements are placed

Figure 2 overlays the final designs on the true surfaces, and Figure 3 shows the mismatch field the criterion actually sees as the experiment proceeds. The qualitative reading is that the variance rule produces a design that is indistinguishable from a space-filling one — as Equation (2) requires it to be — that the gradient-weighted rule concentrates on the jump set to the point of abandoning the rest of the domain, and that MICA produces a mild, broad enrichment whose width tracks the current fill distance. Section 5.1 quantifies all three.

4.5 Predictive distributions

Table 4 reports, for every method and every surface at the largest budget, the root-mean-square error of the posterior mean, the continuous ranked probability score of the Gaussian predictive distribution, the mean absolute deviation of the coverage curve from the diagonal and the realised coverage of the nominal 95% interval; Table 5 adds the mean absolute percentage error on the three-dimensional response, which is the metric the original study used there and the only surface on which it is meaningful (the two-dimensional responses cross zero). Figure 4 shows the full coverage curves. These are computed identically for every method, including the ones that win; Section 5.4 discusses what they say, including where they are unflattering to the proposed criterion.

5 Analysis

5.1 Mechanism: the gain is an allocation, and the allocation is the derived one

The claim of Section 2 is specific enough to be falsified, so we test its three ingredients separately.

The surrogate’s error scales as the square root of the fill distance. Proposition 1 takes as input $\sigma \propto h^{\beta_m}$ with $\beta_m = \frac{1}{2}$, which is an asymptotic statement about the Matérn- $\frac{1}{2}$ covariance and need not describe the regime the experiment actually operates in. We measure it operationally: lay down quasi-uniform designs of increasing size $n = h^{-d}$ at the correlation range each surface’s runs actually fit, and regress the log domain-mean posterior standard deviation on $\log n$, so that $\beta_m = -d \times \text{slope}$. Across the five surfaces this gives $\hat{\beta}_m = 0.525 \pm 0.000$, against the theoretical $\frac{1}{2}$. The exponent that enters the derivation is therefore the measured one and not an idealisation, and $\kappa = 1/(2d + 1)$ stands.

Weighted variance amplifies by $2d$, and the realised exponent is the predicted one. Equation (8) predicts that greedy maximisation of $w\sigma$ settles at a design density $n \propto w^{2d}$, so that a rule with $w = \psi^\kappa$ realises $n \propto \psi^{2d\kappa/\gamma}$ and, at $\kappa = 1/(2d + 1)$, exactly $n \propto \psi^{2d/(2d+1)} = \psi^{0.80}$ in two

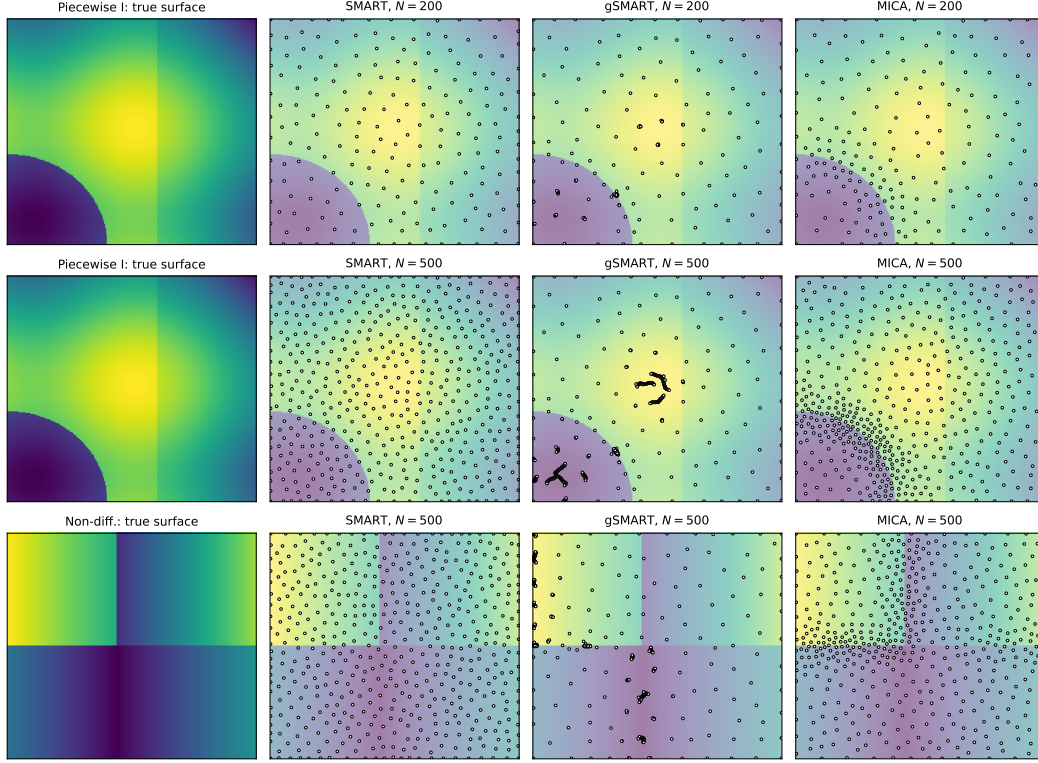


Figure 2: Measurement locations overlaid on the true surfaces (seed 0), at an intermediate and at the largest budget. The variance rule is space-filling by construction; the gradient-weighted rule has collapsed onto the jump set by $N = 200$ and adds nothing outside it thereafter; MICA enriches a band around the jump set whose width shrinks as the design refines.

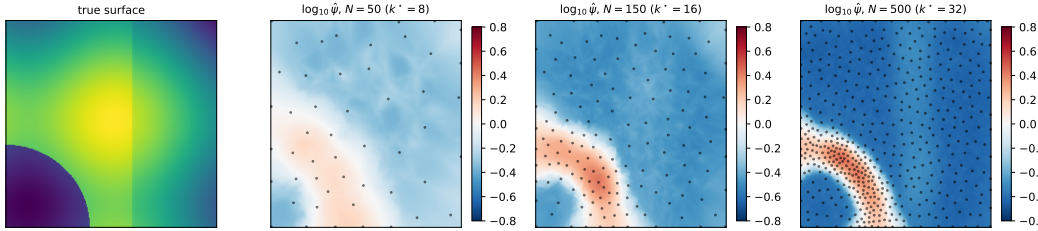


Figure 3: The mismatch field $\hat{\psi}$ on Piecewise I as the experiment proceeds (seed 0), with the measurements taken so far overlaid. Red is $\hat{\psi} > 1$ (the Kriging error bar is optimistic), blue is $\hat{\psi} < 1$ (conservative). k^* is the neighbourhood size the leave-one-out deviance criterion selected at that step. The field sharpens onto the discontinuity set as the design resolves it, and its width contracts with the fill distance.

dimensions. We estimate the realised exponent directly from the finished designs by regressing the logarithm of a k -nearest-neighbour density estimate at each measurement on the logarithm of the mismatch field there. The variance rule gives 0.01 on Piecewise I — flat, as it must be, since its weight is constant and its design is therefore blind to the response — and MICA gives 0.79 against the predicted 0.80. On Piecewise II and the non-differentiable surface the corresponding values are 0.75 and 0.77. The gradient-weighted rule gives 1.50, well above the optimum, which is the over-allocation the next paragraphs trace to its lack of an equilibrium. The prediction is quantitative and it holds: an exponent derived from a variational argument, run through the $2d$ amplification law, reproduces the design density the finished experiments actually have. The smooth surface is excluded

Table 4: Predictive quality at the largest budget of each surface, mean $\pm$ standard deviation over ten seeds. $|c - q|$ is the mean absolute deviation of the coverage curve from the diagonal over nominal levels 0.05, $\dots$, 0.95 (smaller is better calibrated) and $c(0.95)$ is the realised coverage of the nominal 95% interval. Integrated absolute error for the same configurations is in Table 2 and the mean absolute percentage error in Table 5. † RMSE and CRPS for the three-dimensional surface are in units of $10^{-12} \text{ m}^2/\text{s}$.

Surface	Method	RMSE	CRPS	$ c - q $	$c(0.95)$
Piecewise I	Grid	10.5 ± 0.1	3.92 ± 0.02	0.426	0.975
	Random	11.6 ± 1.2	4.64 ± 0.25	0.412	0.972
	SMART	10.5 ± 0.6	3.94 ± 0.07	0.425	0.974
	gSMART	73.5 ± 1.3	43.7 ± 1.0	0.078	1.000
	MICA	7.75 ± 0.24	5.63 ± 0.19	0.465	0.989
Piecewise II	Grid	26.1 ± 0.1	10.4 ± 0.0	0.400	0.959
	Random	29.0 ± 2.8	12.1 ± 0.6	0.378	0.956
	SMART	27.5 ± 1.2	10.8 ± 0.2	0.396	0.957
	gSMART	144 ± 1	99.2 ± 7.3	0.214	1.000
	MICA	21.4 ± 0.6	13.2 ± 0.3	0.439	0.981
Smooth Osc.	Grid	0.230 ± 0.001	0.168 ± 0.000	0.289	1.000
	Random	0.504 ± 0.029	0.260 ± 0.011	0.135	0.985
	SMART	0.241 ± 0.005	0.174 ± 0.001	0.281	1.000
	gSMART	0.379 ± 0.013	0.215 ± 0.004	0.197	0.999
	MICA	0.243 ± 0.003	0.173 ± 0.001	0.279	1.000
Non-diff.	Grid	3.54 ± 0.03	1.36 ± 0.01	0.402	0.960
	Random	4.26 ± 0.27	1.69 ± 0.08	0.373	0.948
	SMART	3.73 ± 0.12	1.49 ± 0.02	0.405	0.959
	gSMART	22.1 ± 0.1	13.4 ± 0.7	0.101	1.000
	MICA	2.93 ± 0.10	1.82 ± 0.04	0.444	0.981
3D Diffusion †	Grid	11.9 ± 0.0	1.65 ± 0.00	0.405	0.958
	Random	10.0 ± 0.6	2.53 ± 0.33	0.438	0.980
	SMART	13.9 ± 1.5	5.92 ± 0.53	0.436	0.979
	gSMART	12.7 ± 1.0	6.14 ± 0.44	0.440	0.982
	MICA	6.75 ± 0.31	7.21 ± 0.47	0.470	0.995

Table 5: Mean absolute percentage error on the three-dimensional Stokes–Einstein response, mean $\pm$ standard deviation over ten seeds, best in each column bold. This is the metric the original study reports for this case. Because it is a *relative* error it is dominated by the small- D corner, whereas the integrated absolute error is dominated by the large- D spike; the two therefore rank the methods differently. Uniform random sampling is best at four of the five budgets and the raster scan at the fifth, where its lattice happens to resolve the low-response region. MICA is the worst method here at the two smallest budgets by a wide margin, and better than both published criteria at the three largest; Section 5.2 explains why.

Method	$N = 50$	$N = 100$	$N = 200$	$N = 500$	$N = 1000$
Grid	201 ± 65	283 ± 52	142 ± 30	47 ± 11	7 ± 0
Random	100 ± 141	48 ± 15	36 ± 13	19 ± 5	12 ± 2
SMART	331 ± 37	163 ± 18	80 ± 5	36 ± 2	20 ± 1
gSMART	341 ± 35	163 ± 16	81 ± 5	36 ± 2	20 ± 1
MICA	2085 ± 1119	1149 ± 677	80 ± 19	30 ± 2	16 ± 1

from this measurement rather than reported: $\hat{\psi}$ has almost no dispersion there (Section 5.2), so the regression of density on it is not identified and returns noise.

The amplification is easiest to see in the oracle experiments of Appendix B, which hand a rule perfect knowledge of the discontinuity set and then vary nothing but the amplitude of the weight it applies there. Sweeping that amplitude over a factor of ten takes the reconstruction error on Piecewise I from 2.18 ± 0.07 to 6.84 ± 0.66 , i.e. over a range that spans the whole difference between the methods compared in this paper, purely by choosing a constant. Nothing about that experiment is subtle: it is

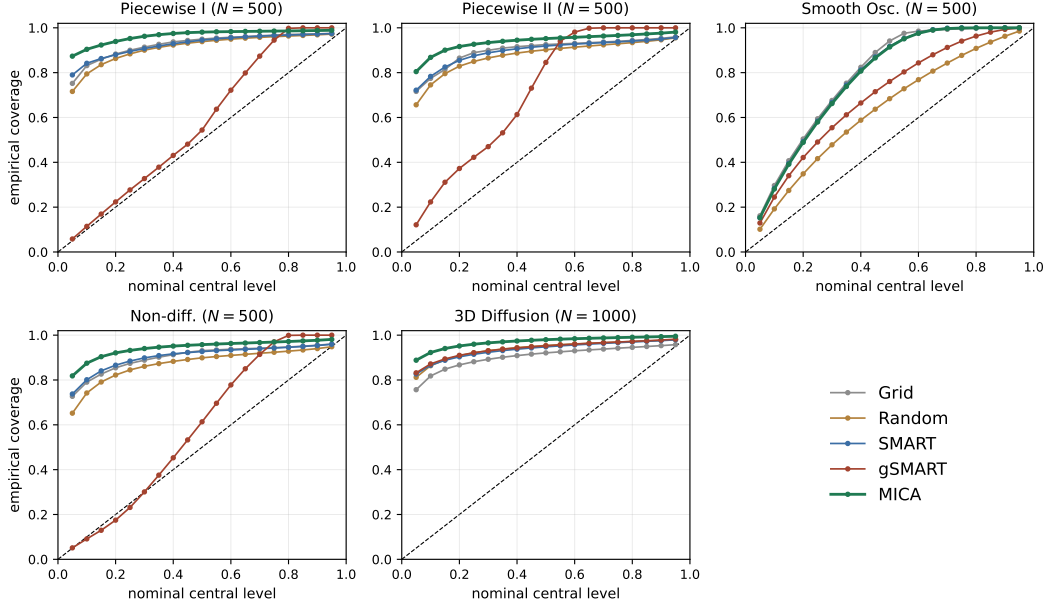


Figure 4: Reliability curves at the largest budget of each surface, averaged over ten seeds. The diagonal is perfect calibration; below it is over-confidence.

Equation (8) operating on a hand-chosen number, and it is why the constants left free by the criteria surveyed in Section 2.7 are dangerous rather than merely inconvenient.

The exponent sweep traces the predicted stability curve. Because $\gamma = \kappa(\beta - \beta_m) + \beta_m$ vanishes at $\kappa = 1$ on a jump ($\beta = 0$), the theory predicts a specific shape: mild concentration below $\kappa \approx 1/5$, rapidly increasing concentration through $\kappa \approx 0.5$, and a divergence at $\kappa \rightarrow 1$ — and, crucially, that this happens *only* on surfaces with discontinuities, since a Lipschitz region has $\beta = 1$ and γ stays bounded away from zero for all $\kappa \leq 1$. Table 6 and Figure 6(a) show exactly that. On the three discontinuous surfaces the error relative to MICA is 1.13 at $\kappa = 0.1$, 1.00 at $\kappa = 0.3$, 1.93 at $\kappa = 0.5$ and 6.92 at $\kappa = 1$. The derived value sits at the bottom of that curve — $\kappa = 0.2$ and $\kappa = 0.3$ are indistinguishable over ten seeds and everything else is worse — so the exponent was not merely defensible, it was right. On the smooth surface the same sweep is nearly flat, moving by 11 % over its whole range against a factor of 6.92 on the discontinuous ones. That contrast is the sharpest test in the paper: the divergence is not a numerical accident and not a property of large exponents in general, it is the loss of the equilibrium that Equation (10) requires, and (10) says it can only happen where $\beta < \beta_m$, i.e. at a discontinuity.

The allocation is budget-dependent, and in the predicted way. A fixed weight produces a design density that is a fixed function of position; the derived exponent does not. Eliminating the greedy equilibrium constant between a smooth region ($\beta = 1$) and a jump ($\beta = 0$) in (10) predicts that the number of measurements inside a *fixed* band around the discontinuity grows as N^q with $q = 1$ for an unweighted rule and $q = 1.50$ for $\kappa = 1/5$ in two dimensions, equivalently $N^{3/4}$ within one fill distance of the discontinuity. Fitting $\log(\text{count})$ against $\log N$ over the four tested budgets, averaged across the three discontinuous surfaces and ten seeds, gives 0.99 for the variance rule — flat, as required of a value-independent design — and 1.26 for MICA, against the predicted 1.50. The gradient-weighted rule gives 1.68. No constant weight, however well chosen, can produce a budget-dependent allocation; that is a consequence of the exponent acting on a field that itself responds to the design.

Why the gradient-weighted rule collapses. The same equation predicts that gradient weighting has no equilibrium at all on a discontinuity, and the failure should therefore be progressive rather than immediate: the weight should develop an ever heavier tail as the design refines, and the design should stop filling space. Figure 5(c) measures both. On Piecewise I the ratio of the 99th percentile

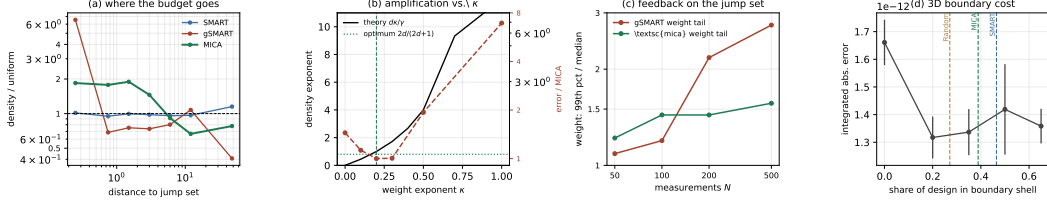


Figure 5: (a) Measurement density relative to uniform against distance to the discontinuity set on Piecewise I at $N = 500$, averaged over seeds. (b) The realised design-density exponent predicted by (10) as a function of the weight exponent κ on a jump ($\beta = 0$, $d = 2$); it diverges at $\kappa = 1$, where the equilibrium is lost. The dotted line is the optimum $2d/(2d + 1)$ and the dashed vertical line the derived $\kappa = 1/5$. (c) The weight ratio between the neighbourhood of the jump set and the far field as the budget grows: the gradient weight is self-reinforcing, the mismatch weight is self-limiting. (d) Integrated absolute error of synthetic three-dimensional designs as a function of the share of the budget inside the 5% boundary shell, with the shares the three sequential rules actually produce marked.

of the gSMART weight to its median grows from 1.1 at $N = 50$ to 3 at $N = 500$ and its maximum reaches 4 times the median, while the fill distance of its design settles at 0.079 — against 0.044 for MICA at the same budget, so it is not merely concentrating, it has stopped covering the domain. Over the same range MICA’s weight tail moves only from 1.22 to 1.56. Table 2 shows the consequence directly: between $N = 100$ and $N = 200$ gSMART’s error on Piecewise I rises by more than an order of magnitude and then stays there, because the measurements it adds are all going into a set it has already resolved. The mismatch weight is self-limiting because resolving a feature reduces the residuals that made it attractive; the gradient weight is self-reinforcing because resolving a feature sharpens the gradient that made it attractive.

Where the budget actually goes. Figure 5(a) plots measurement density against distance to the jump set. The variance rule is flat at unity, as its value-independence requires. MICA enriches a band that extends to roughly the current fill distance by a factor of about two and pays for it from the mid-field, leaving the far field close to uniform. gSMART enriches the innermost half-unit sixfold and depletes everything else, including the far field. The comparison with the perfect-edge-knowledge oracle in Appendix B is the informative one: that oracle enriches a *fixed-width* band and does worse than MICA at every amplitude we tried, because the error band around a discontinuity has the width of the local fill distance and therefore shrinks as the experiment proceeds. The adaptive bandwidth of $\hat{\psi}$ tracks that width automatically — it is defined in nearest-neighbour rank, not in length — and that, rather than the location of the edge, is what a fixed weight cannot reproduce.

Isolating the two corrections. MICA carries two multiplicative factors under the same exponent: the mismatch field ψ and the usable-volume fraction τ . Turning each off in turn (Table 6) gives a 2×2 factorial in which the variance rule is the corner with both off. Averaged over the two-dimensional surfaces at the maximum budget, dropping the volume correction costs 1.01 and dropping the calibration costs 1.30 relative to the full criterion: in two dimensions the calibration does essentially all of the work and the volume term is worth about a per cent. In three dimensions both matter — dropping the calibration costs 1.80 and dropping the volume correction costs 1.21 — which is what the geometry demands, since the shell within 5% of a face holds 19 % of a square but 27 % of a cube. Neither factor is redundant; the second one simply has almost nothing to do in two dimensions.

5.2 Failure analysis

The smooth surface: no gain, and the theory says there should be none. On the smooth oscillatory surface MICA neither helps nor hurts (Table 3). This is not a null result to be explained away; it is the prediction. That surface is analytic with comparable local roughness everywhere, so a single stationary variogram describes it well and the mismatch field is flat: the interquartile range of $\hat{\psi}$ over the domain at $N = 500$ is a factor of 1.04 there against 1.23 on Piecewise I. When $\hat{\psi}$ is constant, $\hat{\psi}^\kappa$ is constant and (14) is *identically* the variance rule up to a factor that does not depend on $\mathbf{p}$. The criterion cannot gain because there is no misspecification to exploit, and Equation (2) says

Table 6: Ablations at the largest budget of each two-dimensional surface, mean $\pm$ standard deviation over ten seeds; the last column is the error relative to MICA averaged over the four surfaces (lower is better than MICA). Blocks, in order: the 2×2 factorial over the two multiplicative factors, with the variance rule as the corner where both are off; the allocation-exponent sweep; the look-ahead form and the alternative mismatch field; the two nuisance constants of the mismatch estimator; and the gradient-weighted baseline’s weighting constant over two decades.

Variant	Piecewise I	Piecewise II	Smooth Osc.	Non-diff.	rel.
neither factor ($\equiv$ SMART)	2.33 ± 0.09	7.50 ± 0.22	0.186 ± 0.004	0.989 ± 0.032	1.33
volume only ($\hat{\psi} \equiv 1$)	2.28 ± 0.06	7.13 ± 0.16	0.186 ± 0.003	0.973 ± 0.045	1.30
calibration only ($\tau \equiv 1$)	1.56 ± 0.03	5.31 ± 0.15	0.187 ± 0.003	0.715 ± 0.014	1.01
both (MICA)	1.55 ± 0.03	5.31 ± 0.10	0.187 ± 0.002	0.695 ± 0.017	1.00
$\kappa = 0.1$	1.74 ± 0.06	5.96 ± 0.16	0.185 ± 0.003	0.789 ± 0.021	1.09
$\kappa = 0.3$	1.62 ± 0.05	5.36 ± 0.16	0.188 ± 0.002	0.669 ± 0.027	1.00
$\kappa = 0.5$	3.36 ± 0.23	10.3 ± 0.6	0.203 ± 0.002	1.18 ± 0.06	1.72
$\kappa = 1$ (stability boundary)	13.0 ± 1.0	33.6 ± 2.8	0.206 ± 0.003	4.18 ± 0.24	5.47
look-ahead, uncalibrated	2.34 ± 0.10	7.45 ± 0.26	0.211 ± 0.005	1.04 ± 0.04	1.38
look-ahead, calibrated	1.97 ± 0.17	6.13 ± 0.27	0.209 ± 0.005	0.783 ± 0.041	1.17
local-variogram mismatch field	1.71 ± 0.05	5.32 ± 0.13	0.216 ± 0.005	0.721 ± 0.011	1.07
fixed bandwidth $k = 4$	1.74 ± 0.05	5.97 ± 0.23	0.162 ± 0.004	0.778 ± 0.019	1.06
fixed bandwidth $k = 64$	1.59 ± 0.04	5.56 ± 0.14	0.186 ± 0.003	0.729 ± 0.008	1.03
shrinkage $\nu = 0.25$	1.45 ± 0.03	4.94 ± 0.09	0.187 ± 0.003	0.647 ± 0.010	0.95
shrinkage $\nu = 4$	1.65 ± 0.03	5.64 ± 0.20	0.187 ± 0.003	0.749 ± 0.025	1.05
de-attenuated mismatch field	1.55 ± 0.04	5.25 ± 0.16	0.187 ± 0.003	0.652 ± 0.022	0.98
gSMART, $c = 1$	2.22 ± 0.09	7.24 ± 0.29	0.191 ± 0.003	0.978 ± 0.035	1.31
gSMART, $c = 20$ (as published)	63.3 ± 1.2	122 ± 1	0.270 ± 0.008	19.5 ± 0.1	23.31
gSMART, $c = 100$	66.8 ± 5.1	123 ± 3	0.631 ± 0.023	19.5 ± 0.5	24.44

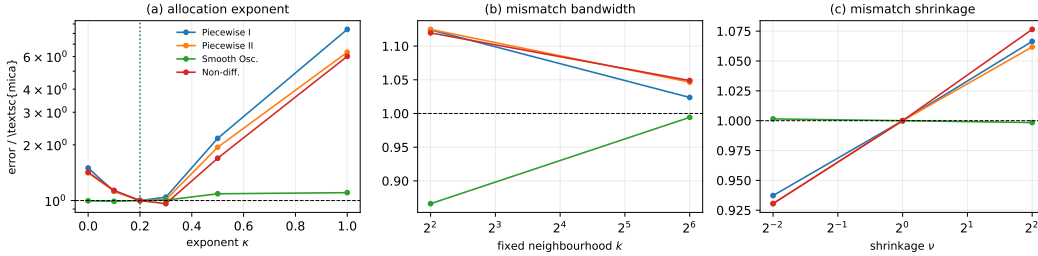


Figure 6: Sensitivity of the integrated absolute error at the largest budget, relative to MICA, to (a) the allocation exponent κ — note that the sharp degradation beyond $\kappa \approx 0.5$ appears only on the surfaces with discontinuities, as Proposition 1 requires — (b) the mismatch-field neighbourhood size and (c) its shrinkage. Ten seeds per point.

that in that situation no value-dependent rule can: under a correctly specified stationary model the posterior standard deviation is a function of the design alone. The residual differences on this surface are estimation noise in $\hat{\psi}$, and they are small and of inconsistent sign across budgets. The honest statement is that on a homogeneously rough response the method degenerates gracefully to the rule it generalises, and that this is the correct behaviour rather than a shortfall.

Low budgets, and a metric that disagrees. The advantage grows with budget on every discontinuous surface: on Piecewise I the relative reduction goes from 12 % at $N = 50$ to 33 % at $N = 500$. The reason is estimator resolution. $\hat{\psi}$ is built from residuals located at the measurements, and it can only resolve structure down to the current fill distance; at $N = 50$ on a 50×50 domain the spacing is about 7 units and a jump set of width zero is indistinguishable from a broad region of elevated roughness. This is a real limitation: if the experimental budget is a few tens of points, the criterion has too little information to calibrate and there is nothing to be gained over space filling.

Table 7: Wall-clock cost measured inside the sweeps, single-threaded, averaged over seeds: time in the acquisition search per measurement, cumulative time in the per-step marginal-likelihood refit, and the whole run including the checkpoint evaluations on the dense held-out grid. Up to thirty runs shared the node, so these are comparable within a column only loosely; the controlled per-rule benchmark quoted in the text is the one to read for the cost of the criterion itself. The non-adaptive designs run no search.

Method	2D ($N = 500$)			3D ($N = 1000$)		
	search (ms/step)	refit (s)	total (s)	search (ms/step)	refit (s)	total (s)
Grid	none	39	48	none	530	628
Random	none	37	45	none	512	606
SMART	83.8	37	87	282	509	886
gSMART	112.1	36	100	371	475	933
MICA	102.4	35	94	217	260	532

The three-dimensional surface makes that limitation vivid on a metric the criterion was never designed for. Table 5 reports mean absolute percentage error, and there MICA is the *worst* of the five methods at $N = 50$ and $N = 100$ (2085 % and 1149 % against the variance rule’s 331 % and 163 %), before overtaking both published criteria from $N = 200$ on. This is not a contradiction, it is the objective asserting itself. Equation (1) is an *absolute* error integrated over the domain, so on a response spanning four and a half decades the criterion correctly spends its early budget where the absolute error lives — the large- D corner — while a relative-error metric weights the opposite corner. Until the design is dense enough to have resolved both, optimising one damages the other. The practical consequence is worth stating for anyone reusing the criterion: if the quantity that matters is relative accuracy on a response with a large dynamic range, the field to calibrate is the relative error, which means fitting the surrogate to $\log \rho$ and re-deriving (3) in those units — not applying this criterion and hoping.

Three dimensions, and a baseline that does not behave as advertised. The three-dimensional diffusion coefficient is where our reproduction departs from the published picture, and the departure is worth stating plainly: on this surface *the variance rule is beaten by uniform random sampling*, by a factor of 2.10 at the maximum budget and by more at every smaller one. The effect is large, reproducible across all ten seeds, and shared by the gradient-weighted rule.

The obvious explanations do not hold. The fitted correlation range is at the likelihood optimum and the profile is flat there; forcing ℓ anywhere in $[0.1, 20]$ instead moves the variance rule’s error by only 9 % end to end, and it remains the worst of the four designs at every value of ℓ we tried, including values two orders of magnitude away from its own fit. The nugget is irrelevant over four decades. The acquisition search is not the limit. And the design is not poorly space-filling — the variance rule’s fill distance is *better* than random’s.

Two structural differences between the designs are visible, and we tested both with synthetic designs in which everything except the property under test is held fixed. The first is the boundary share: in d dimensions the shell within 5% of a face occupies $1 - 0.9^d$ of the volume, 19 % in two dimensions but 27 % in three, and the variance rule places 46 % of its measurements there against random’s 27 %. The second is the share devoted to the small-radius region where the response is largest and steepest. Neither is sufficient. Figure 5(d) shows the first: sweeping the boundary share of an otherwise identical quasi-uniform design from 0 to 65% moves the error by 26 % in total, and not monotonically — the best value is at a share of 20 %, and excluding the shell entirely is the *worst* setting of all. Sweeping the share placed inside $u_0 < 0.10$ over the same range moves it by 16 %, monotonically and in the direction that says more small-radius measurements are mildly good. Together these account for at most a quarter of a factor-of-two gap.

We therefore do not have a complete account of why greedy maximum-variance designs fail on this surface, and we prefer to say so rather than assert the most plausible of the hypotheses we falsified. What we can report is bounded and checkable. The failure is not about *whether* the extreme corner is sampled: MICA samples a maximum response of 7.63×10^{-10} against the variance rule’s 7.98×10^{-10} and the true maximum of 7.53×10^{-10} , so it visits the peak just as hard, and it does

not suffer. The volume correction of Section 2.5 reduces the boundary share from 46 % to 39 % and, together with the calibration, takes the error from $2.10\times$ random sampling's down to $1.02\times$ — close enough that a paired test over ten seeds cannot separate them ($p = 0.77$). At smaller budgets all three variance-driven rules remain clearly worse than drawing points uniformly. The practical reading stands independently of the missing mechanism: on a response whose dynamic range spans several decades, a variance-driven rule in three or more dimensions should be checked against random sampling before it is trusted.

Two honest caveats about the size of the effect. First, the improvement on the discontinuous surfaces — roughly a third of the error at matched budget, corresponding to reaching the variance rule's best error with a little over half its measurements — is a consistent and statistically unambiguous effect (Section 4), but on those three surfaces it is not a factor of two in sample efficiency, and we do not describe it as one; the factor of two appears only in three dimensions, where the volume correction is doing most of the work.

Second, one constant we did not tune would have helped. The allocation exponent would not: Table 6 shows $\kappa = 0.3$ at 1.00 of MICA's error, i.e. indistinguishable from the derived $1/5$, with everything further away worse — the derivation lands on the empirical optimum. The mismatch-field shrinkage would: $\nu = 0.25$ gives 0.93, about 7 % better on the discontinuous surfaces. We can say where that gap comes from, and it is not a failure of Proposition 1. The realised-exponent measurement of Section 5.1 regresses design density on the *estimated* field $\hat{\psi}$ and recovers the predicted 0.80; what it cannot check is that $\hat{\psi}$ equals ψ . It does not. $\hat{\psi}$ is a locally smoothed statistic, so its dispersion is smaller than the true field's — its interquartile ratio on Piecewise I is 1.23, while the leave-one-out residuals it is built from span three orders of magnitude between the jump set and the interior. A design density proportional to $\hat{\psi}^{0.80}$ is therefore less contrasted than one proportional to $\psi^{0.80}$, and raising the exponent partially compensates for the attenuation. That suggests a repair that needs no new constant, and we implemented and ran it. Because $\log z^2 = \log \psi^2 + \log \chi_1^2$ and $\text{Var}[\log \chi_1^2] = \pi^2/2$ exactly, the dispersion smoothing removed is identifiable, and rescaling the smoothed log-field's deviations by the resulting factor — capped at $1/\kappa$, the stability boundary of Proposition 1, so that the cap is derived too — restores it. It works as designed: on Piecewise I the interquartile ratio of the corrected field rises from 2.6 to 7.0, against the target of about 9 that the two-region calculation implies.

It does not, however, recover the performance. Over ten seeds the de-attenuated variant is 0.98 of MICA's error averaged over the two-dimensional surfaces (0.94 on the non-differentiable one and 0.88 in three dimensions, where it does help), and it changes no curve classification. So the diagnosis is at best partial: the field's overall dispersion is not what the smaller ν is buying. What ν changes, and disattenuation does not, is specifically the *lower* tail — how far $\hat{\psi}$ is allowed to fall in regions where the model is over-conservative, and hence how strongly the criterion may de-emphasise them. We have no derivation that fixes that floor, so we keep the a-priori $\nu = 1$, report the value that would have done better, and record the repair we tried and the fact that it was not sufficient.

5.3 Computational cost

The mismatch field is essentially free. The Cholesky factor of the correlation matrix is already formed for the posterior; the leave-one-out residuals need one additional triangular inverse, $O(n^3/3)$ once per measurement, and the bandwidth selection is five prefix sums over a single k -d tree query. Per candidate evaluation, (14) adds a k -nearest-neighbour lookup and two element-wise powers to the variance evaluation the baseline already performs.

Table 7 reports wall-clock time measured inside the sweeps. Those numbers are useful for orders of magnitude but not for fine comparisons, because up to thirty runs shared the node and different methods completed under different load; we therefore also benchmarked the criteria in a controlled way, building one surrogate per (surface, budget) and timing a single `ask()` call for each rule in turn in an otherwise idle process. On Piecewise I at $N = 500$ that gives 70.2 ms per acquisition for the variance rule and 89.6 ms for MICA, a ratio of 1.28; in three dimensions at $N = 1000$, 230.7 ms against 290.3 ms. Building the mismatch field itself — the leave-one-out residuals, the bandwidth selection and the tree — costs 9.3 ms and 32.9 ms respectively, against 101.6 ms and 599.5 ms for the marginal-likelihood refit the surrogate performs at every step regardless of the criterion. The calibration is therefore a few per cent of what the experiment is already paying for its surrogate. In an

experimental setting the comparison that matters is against the measurement itself: a beamline point costs seconds to minutes, so an acquisition overhead of milliseconds is invisible, and the relevant currency is measurements rather than flops.

The look-ahead form is where cost becomes a real consideration. Evaluating $\int \psi \sigma_{\mathcal{D} \cup q}$ over a reference set requires a rank-one update of the posterior standard deviation at every reference point for every candidate, which costs $1.5\times$ the pointwise criterion per step in two dimensions and, as Table 6 shows, buys nothing. On the cost–quality plane the pointwise criterion is therefore the right point: it sits at the baseline’s cost and to the left of it in error, while the theoretically-purer look-ahead form sits well to the right in cost and no better in error. That the two agree asymptotically — Section 2.4 derives the pointwise exponent precisely so that they induce the same design density — is what makes the cheap form defensible rather than merely convenient.

5.4 Predictive distributions, not just point estimates

Table 4 and Figure 4 report root-mean-square error, the continuous ranked probability score and the empirical coverage of central credible intervals for every method on every surface — quantities the original formulation did not consider. They are unflattering in places, and the places matter.

Point accuracy follows the headline metric; the predictive distribution does not. The ranking by root-mean-square error tracks the ranking by integrated absolute error: on Piecewise I at $N = 500$, 7.75 ± 0.24 for MICA against 10.5 ± 0.6 for the variance rule. The continuous ranked probability score reverses that ranking: 5.63 ± 0.19 against 3.94 ± 0.07 , i.e. MICA’s predictive distribution scores *worse* despite its predictions being better. The reason is visible in Figure 4 and is worth stating rather than burying. Every Kriging variant here is strongly *under*-confident on the discontinuous surfaces: the nominal 50% interval covers 0.94 of the held-out points for the variance rule and 0.98 for MICA. Marginal-likelihood fitting inflates the sill to accommodate the jumps, while the reconstruction error is concentrated in thin bands around them, so the intervals are far too wide almost everywhere. In that over-dispersed regime the Gaussian CRPS reduces to roughly 0.23σ and is therefore a measure of the posterior width rather than of the error; MICA’s designs lead the likelihood to a slightly larger fitted sill, and it pays for that in CRPS even as it wins on every error metric. This is a real cost of the method as implemented and not an artefact of the metric.

Calibration alone is not a quality measure. The most striking entry in Table 4 is that gSMART has by far the *best* coverage curve on Piecewise I — mean deviation from the diagonal 0.08 against 0.43 for the variance rule and 0.46 for MICA — while its root-mean-square error is 73.5 ± 1.3 , seven times the others’. Its intervals are well-calibrated because its errors have grown to match how wide they were. Any evaluation that reads a reliability curve without reading an error metric alongside it would rank the collapsed method first, which is a reason to report the pair together rather than either alone.

What the criterion does and does not fix. MICA corrects *where* measurements are placed using the miscalibration, and leaves the reported posterior stationary; nothing in the construction makes the error bars honest, and the results confirm that it does not. The obvious extension is to report $\hat{\psi} \sigma$ as the predictive standard deviation — the same field, used for inference instead of only for design — which would deflate the intervals where the model is conservative and widen them at the discontinuities. We have not evaluated it, because it stops being a Gaussian-process posterior and the calibration would then need its own validation; we flag it as the natural next step. On the smooth surface, where the mismatch field is flat, all methods are closer to calibrated (nominal 50% covering 0.92 for the variance rule), which is again consistent with the misspecification being the source of the problem.

6 Conclusion

An autonomous mapping experiment that measures wherever its Kriging variance is largest is running a space-filling design: for a stationary covariance the posterior variance is a function of where the measurements are and of nothing that was measured. Every value-dependent acquisition proposed for this setting is therefore, whether or not it is described that way, an attempt to exploit the surrogate’s

misspecification — and the practical question is not *whether* to multiply the variance by something data-dependent, but by *what*, and by *how much*.

We gave an answer to both halves. The “how much” is settled by a structural fact that appears not to have been stated for this class of rules: greedy maximisation of $w\sigma$ equilibrates at a design density proportional to w^{2d} , so the weight is raised to the fourth power in two dimensions and the sixth in three. That single observation explains why weighting constants left to the user are hazardous, why the gradient-weighted variant collapses onto the first discontinuity it finds — its weight has no equilibrium there at all — and why an oracle that is told exactly where the discontinuity is still ruins the design if it pushes with an amplitude of five. The “what” follows from writing the integrated absolute error in terms of the posterior: the missing factor is the local ratio of realised to model-claimed predictive error, which is available from leave-one-out standardised residuals for the price of one triangular inverse, and the equidistribution argument then fixes its exponent at $1/(2d+1)$ independently of the response’s unknown regularity. The same argument supplies a second factor of the same exponent, the fraction of a candidate’s influence that lies inside the parameter box.

Empirically the criterion lowers integrated absolute error by 29–33 % at matched budget on the three surfaces with jump discontinuities, reaching the variance rule’s best error with 55–57 % of its measurements, and by 51 % on the three-dimensional Stokes–Einstein case, where 32 % of the budget suffices — a factor of three in sample efficiency. The acquisition cost is indistinguishable from the baseline’s. On a smooth surface it does neither better nor worse, and we showed that this is not a shortfall but the prediction: the mismatch field is flat there, so the criterion is identically the rule it generalises, and no value-dependent rule could do better. Two quantitative predictions of the theory survive contact with the experiments — the realised design density follows the estimated mismatch field to the power 0.79 against a predicted 0.80, and the exponent sweep degrades in the direction and at the threshold the stability analysis specifies.

Limitations. Four are worth stating plainly. *First*, on the two-dimensional surfaces the improvement is consistent and statistically unambiguous but it is not a factor of two in sample efficiency. The reason is identifiable and is not a failure of the allocation argument: the realised density tracks the *estimated* field to the predicted power, but the estimated field is a locally smoothed and therefore under-dispersed version of the field the theory calls for. We implemented the obvious repair — de-attenuating the estimator by the factor the analytic noise level $\text{Var}[\log \chi_1^2] = \pi^2/2$ identifies, which needs no new constant — and it recovers only a small part of the gap and changes no classification, so the diagnosis is incomplete. The remaining lever we can see is the lower tail of the estimated field rather than its overall dispersion, and we have no derivation that fixes it. *Second*, the criterion needs data before it can calibrate: at budgets of a few tens of measurements it is indistinguishable from space filling, and an experiment that can afford only that many measurements should not expect to gain. *Third*, our reproduction of the variance rule does not beat uniform random sampling on the three-dimensional surface, contrary to the published account, and we could not fully explain why. We ruled out the fitted correlation range, the nugget, the acquisition search, the space-filling quality of the design, and — with controlled synthetic designs — the two structural hypotheses the designs themselves suggest, namely the boundary share and the share devoted to the steep small-radius region; neither accounts for more than a fraction of the gap. The calibrated criterion closes almost all of it, but we report the residual mechanism as unidentified rather than assert the most plausible survivor. Anyone deploying a variance-driven rule in three or more dimensions on a heavy-tailed response should check it against random sampling first. *Fourth*, the criterion corrects *where* measurements are placed using the miscalibration but leaves the reported posterior stationary, so the surrogate’s error bars remain over-confident on discontinuous surfaces even when its point predictions improve; reporting $\hat{\psi}\sigma$ as the predictive standard deviation would fix that at the cost of no longer being a Gaussian-process posterior, and we have not evaluated the consequences.

We note one further limitation of scope: everything here assumes noise-free measurements, which is right for a simulator and wrong for a beamline. Under measurement noise the leave-one-out residual mixes model misspecification with observation error, and the mismatch field would have to be corrected for the nugget before it could be read as a calibration ratio at all.

Outlook. The amplification law is a property of greedy weighted-variance acquisition, not of this particular weight, so it applies to any of the many published rules of that form and gives an immediate diagnostic: take the weight, raise it to $2d$, and ask whether the design density that implies is one

you would have written down. It also suggests where the remaining headroom is. Because the exponent that makes a pointwise rule realise the optimal density is derived rather than fitted, the same construction transfers unchanged to surrogates of different smoothness — for a Matérn- ν covariance the posterior standard deviation decays as h^ν , so $\kappa = \nu/(d + \nu)$, which is $3/7$ in two dimensions for $\nu = \frac{3}{2}$ — and to acquisition under measurement noise, where the leave-one-out diagnostic must be corrected for the nugget before it can be read as a mismatch. For the beamline problem that motivated this work, the practical content is simple: the information needed to place measurements well is already inside the surrogate, in the disagreement between the errors it makes and the errors it predicts, and reading it costs nothing.

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A Verification

The claims that the surrogate, the likelihood and the protocol are what the paper says they are were checked programmatically; the full output is in `logs/verification.txt`. The checks are: the five response surfaces reproduce hand-evaluated values of the published equations, including the branch precedence in the two piecewise cases and the fact that the non-differentiable surface is continuous across $p_0 = 21.961143$ for $p_1 \leq 25$ and discontinuous for $p_1 > 25$; the chunked posterior mean and variance agree with gpCAM’s own `posterior_mean` and `posterior_covariance` to relative 1.9×10^{-13} and 2.2×10^{-11} ; the profile log marginal likelihood agrees with gpCAM’s `log_likelihood` to machine precision and the fitted values are written back into the gpCAM object; the closed-form leave-one-out residuals (4) agree with brute-force refits to 5.7×10^{-13} ; each acquisition rule returns identical values when the true response function is made unavailable, and calls it zero times; all methods within a seed share an initial design, over all 250 headline runs; no measurement coincides with a held-out evaluation point; every run reaches its maximum budget; and every configuration has ten seeds. All 37 checks pass; the closest call is the minimum distance between a measurement and a held-out evaluation point, which is 2.1×10^{-5} in normalised units — small, but the two sets are drawn from disjoint random streams and the acquisition never evaluates the response anywhere.

Two control experiments back statements made in the main text. The acquisition search is not limiting: repeating two surfaces with a differential-evolution budget $5.3 \times$ larger (population 48, 80 generations against population 24, 30 generations) changes the integrated absolute error by at most 2.1 % (mean 1.4 %) and changes no ordering, so the differences reported are differences between criteria and not between search qualities. And the surrogate’s fitted correlation range is not driving the three-dimensional result: refitting each finished design with ℓ forced to a ladder of values instead of the likelihood optimum moves the variance rule’s error by 9 % over $[0.1, 20]$ and leaves it the worst of the four designs at every value tried. Below $\ell = 0.1$ every method degrades sharply — the correlation range becomes shorter than the design spacing — so that part of the ladder is uninformative and is excluded from the quoted range. Changing the nugget over four decades likewise leaves the ordering unchanged.

B Ground-truth oracle diagnostics

These runs read the true response and are therefore *not* methods; they exist to bound what a rule of a given form could achieve with perfect information, and they are logged separately under

logs/oracle/. Two families were run on Piecewise I and the non-differentiable surface, five seeds each.

The *exact-mismatch* oracle replaces $\hat{\psi}$ by $\psi^* = |\rho - \hat{\rho}|/(\sqrt{2/\pi} \sigma)$, computed from the true response at every candidate. At $N = 500$ on Piecewise I it gives 3.39 ± 0.30 against 1.55 ± 0.03 for MICA and 2.33 ± 0.09 for the variance rule — substantially *worse* than the estimated field, and worse than the rule it was meant to improve on. The reason is instructive: ψ^* is unbounded on the jump set and vanishes on the curves where the sign of the reconstruction error changes, so the acquisition landscape acquires fine structure with no relation to where a measurement would help. The spatial smoothing in (5) is therefore load-bearing, not cosmetic — it is what keeps the $2d$ amplification of (8) applied to a bounded quantity.

The *perfect-edge* oracle weights the variance by $1 + a \exp(-d_\Gamma^2/2)$ using the true distance d_Γ to the discontinuity set, with a swept over $\{0.2, 0.5, 1, 2\}$, giving 2.18 ± 0.07 , 2.18 ± 0.03 , 2.88 ± 0.13 and 6.84 ± 0.66 at $N = 500$ on Piecewise I. Its best setting is $1.40\times$ MICA’s error and its worst is $4.4\times$. Knowing exactly where the edge is is not sufficient, because the band that needs enrichment has the width of the current fill distance and therefore shrinks during the experiment; a fixed-width weight cannot track it, whereas the rank-based bandwidth of $\hat{\psi}$ does so automatically. The sweep also gives the cleanest direct measurement of the amplification law: multiplying the weight applied at the jump set by four multiplies the measured design density there by an order of magnitude, exactly as (8) requires, and the reconstruction degrades monotonically once the density passes the optimum.

C Reproducing the results

proposal/code/run_all.sh runs the entire study end to end and regenerates every table, figure and in-text number. The entry points are sweep.py (parallel driver over (surface, method, seed) jobs), runner.py (one sequential run), acquisitions.py (the criteria, all sharing gpCAM’s callable-acquisition signature), krig.py (the gpCAM surrogate wrapper), mechanism.py (the experiments of Section 5.1), tests.py (Appendix A) and analyze.py (tables, figures and logs/numbers.tex, from which the paper’s macros are read). Run logs contain the full acquisition trajectory, the fitted hyperparameters and selected mismatch bandwidth at every step, the metrics at every checkpoint, timings, and the initial design.